PHYSICS

1) A particle moves along X-axis and its position $x = U(t - 4) + 2a(t - 4)^2$
$t \rightarrow$ time
(A) The initial velocity (at t = 0) is U
(B) The acceleration is 4a
(C) The acceleration is 2a
(D) At t = 4 second, particle is at origin
following statements are correct.

(1) A, B

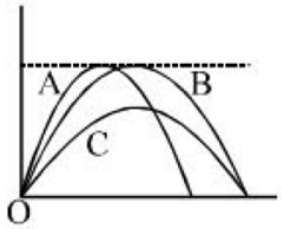
(2) B, C

(3) B, D

(4) A, D

2) Trajectories are shown in figure for three kicked footballs. Initial vertical & horizontal velocity components are u_y and u_x respectively. Ignoring air resistance, choose the correct statement from **Column-II** for the value of variable in **Column-I**.

Column-I		**Column-II**	
(A)	time of flight	(P)	greatest for A only
(B)	u_y/u_x	(Q)	greatest for C only
(C)	u_x	(R)	equal for A and B
(D)	$u_x\ u_y$	(S)	equal for B and C



(1) (A)-S, (B)-R, (C)-P, (D)-Q

(2) (A)-R, (B)-S, (C)-P, (D)-Q

(3) (A)-R, (B)-P, (C)-Q, (D)-S

(4) (A)-R, (B)-S, (C)-Q, (D)-P

3) A motor car covers $\frac{1}{3}$rd part of total distance with $v_1 = 10\,km/hr$, second $\frac{1}{3}$rd part with $v_2 = 20$ km/hr and rest $\frac{1}{3}$rd part with $v_3 = 60\,km/hr$. What is the average speed of the car?

(1) 18 km/hr

(2) 45 km/hr

(3) 6 km/hr

(4) 22.5 km/hr

4) If acceleration of a particle is a = 2x then find out velocity at x = 3 m if particle starts from origin.

(1) $3\sqrt{2}$ m/s

(2) $3\sqrt{3}$ m/s

(3) 2 m/s

(4) $2\sqrt{2}$ m/s

5) The x and y coordinates of a particle at any time t are given by
$x = 7t + 4t^2$ and $y = 5t$
where x and y are in m and t in sec the acceleration of the particle at 5 s is :-

(1) Zero

(2) 8 m/s^2

(3) 20 m/s^2

(4) 40 m/s^2

6) Particle is projected with a velocity of $(10\hat{i} + 12\hat{j})m/s$. If it has a constant acceleration $(3\hat{i} - 3\hat{j})$ of m/s^2. Then select the correct statement :

(1) Initially speed of particle increases

(2) Initially speed of particle decreases

(3) Speed remain constant throughout the motion but its direction changes.

(4) Speed and direction both remain constant.

7) A particle stars from the origin with a velocity of 10m/s and moves with a constant acceleration till the velocity increases to 50m/s. At that instant, the acceleration is suddenly reversed. What will be the velocity of the particle, when it returns to the starting point?

(1) zero

(2) 10m/s

(3) 50m/s

(4) 70m/s

8) The velocity of the bullet becomes one third after it penetrates 4 cm in a wooden block. Assuming that bullet is facing a constant resistance during its motion in the block. The bullet stops completely after travelling at (4 + x) cm inside the block. The value of x is:

(1) 2.0

(2) 1.0

(3) 0. 5

(4) 1.5

9) A car starting from rest accelerates at the rate f through a distance S, then continues at content speed for time 't' and then deaccelerates at the rate f/2 to come to rest. If the total distance travelled is 5S then.

(1) S = ft

(2) $S = \frac{1}{6}ft^2$

(3) $S = \frac{1}{2}ft^2$

(4) $S = \frac{1}{4}ft^2$

10) A stone is dropped from a certain height which can reach the ground in 5 sec. It is stopped after three seconds of its fall and then is again released. The total time taken by the stone to reach the ground will be :-

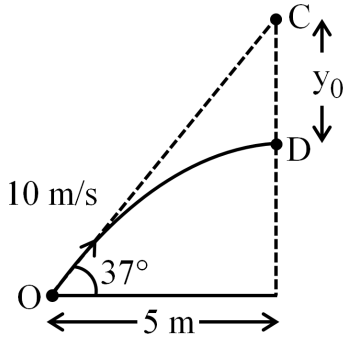
(1) 6 s

(2) 6.5 s

(3) 7 s

(4) 7.5 s

11) A ball is thrown from a point O aiming a target at angle 37° with the horizontal so that the ball hits the target at C but the ball hits at point D, a vertical distance y_0 below C. If the initial velocity of the ball is 10 m/s and the horizontal distance between O and C is 5 m. Find y_0 :-



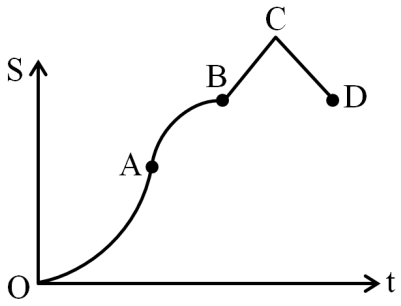
(1) 1 m

(2) 1.5 m

(3) 2.5 m

(4) 1.95 m

12) Displacement-time graph of a particle moving in straight line is as shown in the figure. Choose the incorrect statement :-



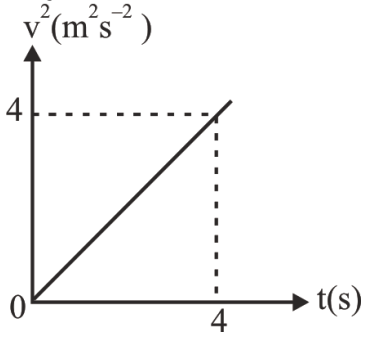
(1) In region OA, acceleration is positive

(2) In region AB, acceleration is negative

(3) In region BC, acceleration is positive

(4) In region CD, acceleration is zero

13) A particle is moving along a straight line such that square of its velocity varies with time as shown in the figure. What is the acceleration of the particle at t = 4 s ?



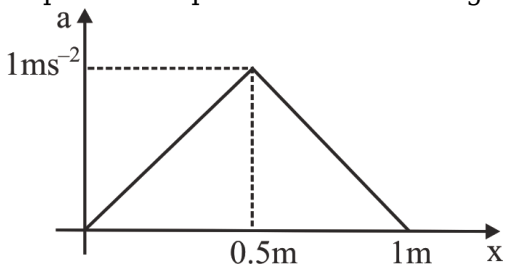
(1) 4 m/s^2

(2) 1/4 m/s^2

(3) 1/2 m/s^2

(4) 0

14) A body initially at rest, starts moving along x-axis in such a way so that its acceleration vs displacement plot is as shown in figure. The maximum velocity of particle is :-



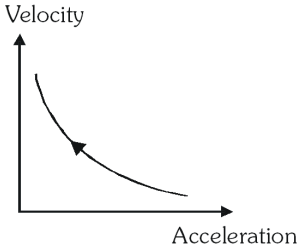
(1) 1 m/s

(2) 6 m/s

(3) 2 m/s

(4) none

15) A physics student studies rectilinear motion of a body and prepares the following graph.



Which of the following conclusions best suits the above graph?

(1) The body is speeding up and its acceleration is decreasing.

(2) The body is slowing down and its acceleration is increasing.

(3) The body is speeding up and its acceleration is increasing.

(4) The body is slowing down and its acceleration is decreasing.

16) A body is projected in such a way that its horizontal range and time of flight both are equal to $\frac{\sqrt{3}}{g}$ in numerical values. The angle of projection will be:-

(1) $\theta = \tan^{-1}\left(\frac{\sqrt{3}}{4}\right)$

(2) $\theta = \tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

(3) $\theta = \tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$

(4) $\theta = \tan^{-1}\left(\frac{5}{8}\right)$

17)

Which of the following quantities remain constant during projectile motion?

(A) Average velocity between two points.

(B) Average speed between two points

(C) $\frac{d\vec{v}}{dt}$

(D) $\frac{d^2}{dt^2}(\vec{v})$

(1) A only

(2) C only

(3) A and C

(4) C and D

18) A ball is thrown from the ground to clear a wall 3 m high at a distance of 6 m and falls 18 m away from the wall, the angle of projection of ball is :-

(1) $\tan^{-1}\left(\frac{3}{2}\right)$

(2) $\tan^{-1}\left(\frac{2}{3}\right)$

(3) $\tan^{-1}\left(\frac{1}{2}\right)$

(4) $\tan^{-1}\left(\frac{3}{4}\right)$

19) A bomber plane moves horizontally with a speed of 500 m/s and a bomb released from it, strikes the ground in 10 sec. Angle at which it strikes the ground with horizontal will be ($g = 10\ m/s^2$)

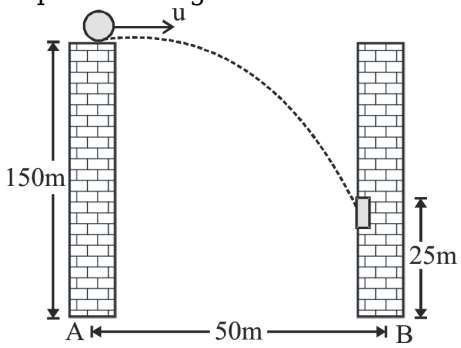
(1) $\tan^{-1}\left(\frac{1}{5}\right)$

(2) $\tan^{-1}\left(\frac{1}{2}\right)$

(3) $\tan^{-1}(1)$

(4) $\tan^{-1}(5)$

20) Two buildings A and B are 50 m apart as shown. With what horizontal velocity 'u' a ball must be thrown from top of building A so that it enters into window on building B, 25 m above the ground (g = 10 m/s^2) ?



(1) 10 m/s

(2) 15 m/s

(3) 20 m/s

(4) 8 m/s

21) A coin is released inside a lift from a height of 2 m from the floor of the lift. The height of the lift is 10 m. The lift is moving with an acceleration of 9 m/s^2 downwards. The time after which the coin will strike with the floor of lift is : ($g = 10\ m/s^2$)

(1) 4s

(2) 2s

(3) $\frac{4}{\sqrt{21}}$ s

(4) $\frac{2}{\sqrt{11}}$ s

22) A particle is kept at rest at origin another particle starts from (–5, 0) m with a velocity of $4\hat{i}-3\hat{j}$ m/s then their closest distance of approach is–

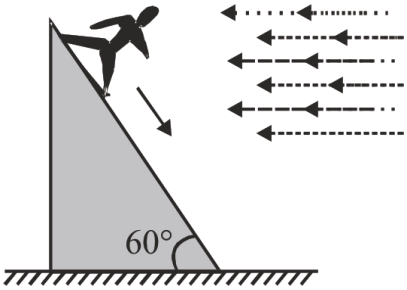
(1) 3 m

(2) 4 m

(3) 5 m

(4) 2 m

23) A man is moving down an incline with a speed v. He meets rain with the same speed but

horizontally. The speed of rain is. 

(1) v

(2) $\sqrt{3}v$

(3) $\frac{\sqrt{3}}{2}v$

(4) $\frac{1}{2}v$

24) A swimmer can swim in still water with speed v and the river flowing with velocity u (u < v). To cross the river in shortest distance, he should swim making angle θ with the upstream. What is the ratio of the time taken to swim across in the shortest time to that in swimming across over shortest distance?

(1) cos θ

(2) sin θ

(3) tan θ

(4) cot θ

25) A boat is traveling east at 12 m/s. A flag flaps at 53° N of W at the bow of the boat. Another flag on shore flaps due north. Then find out the speed (in m/s) of the wind as measured by a stationary observer on land ?

(1) 9 m/s

(2) 12 m/s

(3) 15 m/s

(4) 16 m/s

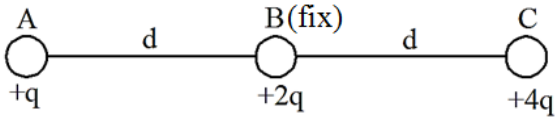
26) Five balls numbered 1 to 5 are suspended using separate threads. Pairs (1, 2), (2, 4) and (4, 1) show electrostatic attraction, while pair (2, 3) and (4, 5) show repulsion. Therefore ball 1 must be

(1) Positively charged

(2) Negatively charged

(3) Neutral

(4) Made of metal

27) Three charges +q, +2q and +4q are connected by strings as shown in the figure. What is ratio of tensions in the strings AB and BC. 

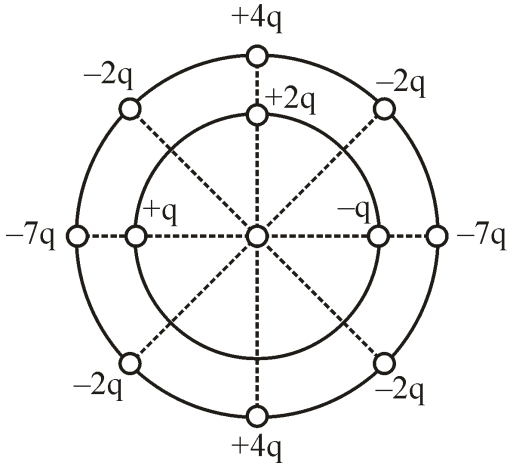
(1) 1 : 2

(2) 1 : 3

(3) 2 : 1

(4) 3 : 1

28) In figure, a central particle of charge –q is surrounded by two circular rings of charged particles of radii r and R, with R > r. What is electrostatic force on the central particle due to the other particles ?



(1) $2\sqrt{2}\frac{kq^2}{r^2}$

(2) Zero

(3) $\frac{kq^2}{r^2}$

(4) $\frac{2kq^2}{r^2}$

29) A particle of mass 0.002 kg and a charge 1 μC is held at rest on a frictionless horizontal surface at a distance of 1m from a fixed charge of 1 mC. If the particle is released, it will be repelled. The speed of the particle when it is at a distance of 10m from the fixed charge is :-

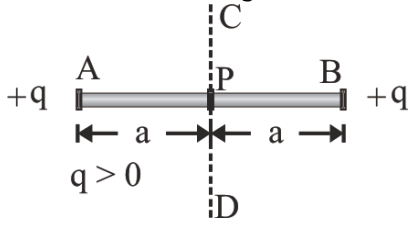
(1) 60 ms^{-1}

(2) 75 ms^{-1}

(3) 90 ms^{-1}

(4) 100 ms^{-1}

30) For the arrangement shown in the figure below, mark out the correct statement.



(1) For positive test charge, P is a position of stable equilibrium for displacement along AB.

(2) For positive test charge, P is a position of unstable equilibrium for displacement along CD.

(3) For –ve test charge, P is a position of unstable equilibrium for displacement along AB.

(4) All of the above

31) The potential of electrostatic field is given by $\phi = xy$, where x & y are coordinate. Which of the following is correct :-

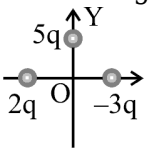
(1) Electric field at all points on positive x-axis is in positive y-direction.

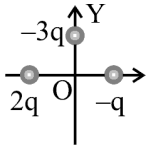
(2) Electric field at all point of positive y-axis is in negative x-direction.

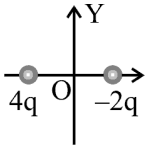
(3) Electric field at (2, 2) is $\sqrt{2}$N/m

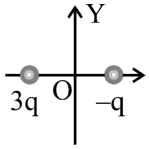
(4) Electric field at x, y makes an angle $\theta = \tan^{-1}\left(\frac{y}{x}\right)$ with x-axis.

32) In the following four situations charged particles are at equal distance from the origin. Arrange them the magnitude of the net electric field at origin greatest first:-

(I)


(II)


(III)


(IV)


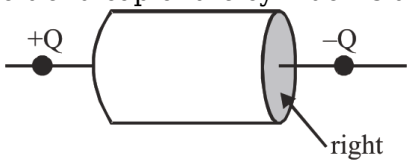
(1) (i) > (ii) > (iii) > (iv)

(2) (ii) > (i) > (iii) > (iv)

(3) (i) > (iii) > (ii) > (iv)

(4) (iv) > (iii) > (ii) > (i)

33) Positive and negative charges of equal magnitude lie along the symmetry axis of a cylinder. The distance from the positive charge to the left end-cap of the cylinder is the same as the distance from



the negative charge to the right end cap. The sign of the flux through the right end-cap of the cylinder is :

(1) Positive

(2) Negative

(3) There is no flux through the right end-cap.

(4) None of these

34) A charged hollow metal sphere has a radius r. if the potential difference between its surface and a point at distance 3r from the centre is V, the the electric intensity at a distance 3r from the centre is:

(1) $\frac{V}{6r}$

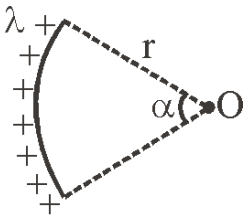
(2) $\frac{V}{4r}$

(3) $\frac{V}{3r}$

(4) $\frac{V}{2r}$

35)

If electric field at centre of charged arc of radius r and linear charge density λ is $\frac{\sqrt{3}\lambda}{4\pi\in_0 r}$. Then α is :

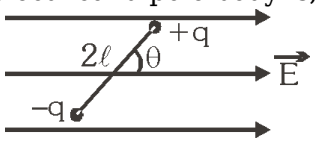


(1) 60°

(2) 120°

(3) 30°

(4) 135°

36) The correct formula for potential energy in the electric field of an electrical dipole body is, when the angle of the electric dipole is θ with the electric field $\left(k=\frac{1}{4\pi\in_0}\right)$: 

(1) U = –pE cos θ

(2) U = +pE cos θ

(3) $U=-pE\cos\theta+\frac{kq^2}{2\ell}$

(4) $U=-pE\cos\theta-\frac{kq^2}{2\ell}$

37) Consider the following statements about electric dipole and select the correct ones:

S_1 : Electric dipole moment vector $\vec{p}$ is directed from the negative charge to positive charge.

S_2 : The electric field of a dipole at a point with position vector $\vec{r}$ depends on $|\vec{r}|$ as well as angle between $\vec{r}$ and $\vec{p}$.

S_3 : The electric dipole potential falls-off as $\frac{1}{r^2}$ and not as $\frac{1}{r}$.

S_4 : In a uniform electric field the electric dipole experiences no net forces but a torque $\vec{\tau}=\vec{p}\times\vec{E}$.

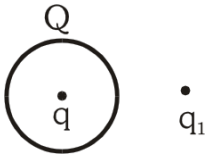
(1) S_2, S_3 and S_4

(2) S_3 and S_4

(3) S_2 and S_3

(4) S_1, S_2, S_3 and S_4

38) A thin, metallic spherical shell contains a charge Q on it. A point charge q is placed at the centre

of the shell and another charge q_1 is placed outside it as shown in figure. All the three charges are positive. The force on the charge at the centre is 

(1) towards left

(2) towards right

(3) upward

(4) zero

39) A spherical charged conductor has σ surface charge density. The electric field on its surface is E. If the radius of the sphere is doubled keeing the surface charge density unchanged, what will be the electric field on the surface of the new sphere:-

(1) $\frac{E}{4}$

(2) $\frac{E}{2}$

(3) E

(4) 2E

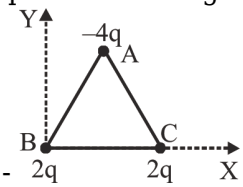
40) Two charged conducting spheres of radii R_1 and R_2 are at same potential then the ratio of their charge densities is:

(1) R_2/R_1

(2) R_1/R_2

(3) R_1^2/R_2^2

(4) R_2^2/R_1^2

41) Charge distribution at three vertices of an equilateral triangle of side a is shown in the figure. The net electric dipole moment of the system is:- 

(1) $\sqrt{3} \times 2qa\hat{j}$

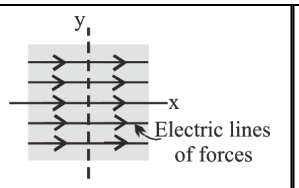
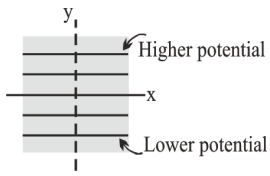
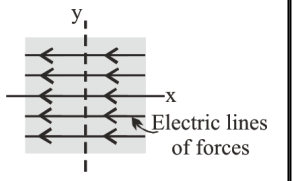
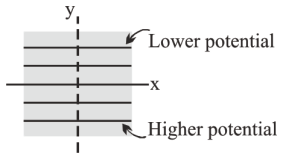
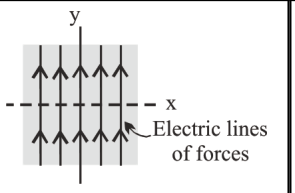
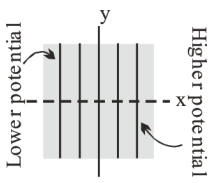
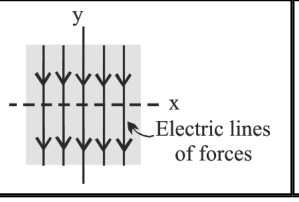
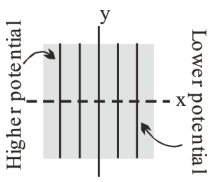
(2) $-2\sqrt{3}qa\hat{j}$

(3) $2\sqrt{3}qa\hat{i} + 2\sqrt{3}qa\hat{j}$

(4) $-2\sqrt{3}qa\hat{i} - 2\sqrt{3}qa\hat{j}$

42) List I gives certain situations in which electric field is represented by electric lines of forces in x-y plane. List II gives corresponding representation of equipotential lines in x-y plane. Match the figures in List I with the figures in List II and indicate your answer.

List - I	**List - II**

(P)		(1)	
(Q)		(2)	
(R)		(3)	
(S)		(4)	

	P	Q	R	S
(1)	1	2	3	4
(2)	4	3	2	1
(3)	3	4	2	1
(4)	2	1	3	4

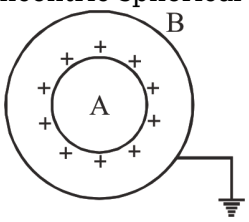
(1) 1

(2) 2

(3) 3

(4) 4

43) A and B are two concentric spherical shells. If A is given a charge +q while B is earthed as shown in figure, then :



(1) Potential on the outer surface of shell B is zero

(2) the charge on B is equal and opposite to that of A

(3) the field inside A and outside B is zero

(4) All of the above

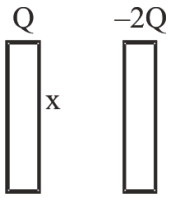
44) ***Assertion (A) :*** Two concentric conducting spherical shells are charged. The charge on the outer shell is varied keeping the charge on inner shell constant, as a result the electric potential difference between the two shells does not change.
Reason (R) : In the situation of Assertion, the electric field in the region between both the shells depends only on the charge existing on the inner shell. :-

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

45) Find the value of x for large metallic plates :- 

(1) –Q

(2) 2Q

(3) $\frac{+3Q}{2}$

(4) 3Q

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	3	1	1	2	2	4	3	3	3	4	3	2	1	1	2	4	2	1	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	1	1	2	4	3	2	1	3	4	2	3	1	1	2	4	4	4	3	1
Q.	41	42	43	44	45															
A.	2	2	4	4	3															

SOLUTIONS

PHYSICS

1) Vel. $V = U + 2a \times 2\,(t - 4)$ $\quad V = U + 4a\,(t - 4)$

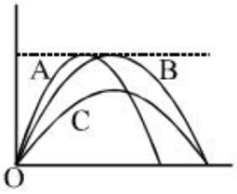
acc^n $= \frac{dv}{dt} = 0 + 4a$

at $t = 0$, $V_0 = U + 4a\,(0 - 4)$

$V_0 = U - 16a$

at $t = 4$, $x_4 = U(4 - 4) + 2a(4 - 4)^2$

$x_4 = 0$

2) 

(A) time of flight (R) equal for A & B

(B) u_y/u_x (P) greater for A Only

(C) u_x (Q) greater for C only because its time is shortest

(D) u_x/u_y (S) equal for B & C because range are equal

3)

$$< v > = \left(\frac{x}{\frac{x}{30} + \frac{x}{60} + \frac{x}{180}}\right) = \left(\frac{1}{\frac{1}{30} + \frac{1}{60} + \frac{1}{180}}\right)$$

$$= \frac{180}{6 + 3 + 1}$$

$< v > = 18$ km / h

4) $$v\frac{dv}{dx} = 2x \int_0^v vdv = \int_0^3 2x\,dx$$

$$\left[\frac{v^2}{2}\right]_0^v = \left[x^2\right]_0^3$$

$v^2 = 2[3^2 - 0]$

5) $x = 7t + 4t^2$

velocity in x direction

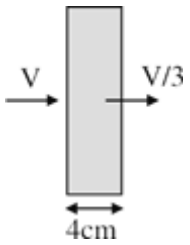
$$v_x = \frac{dx}{dt} = 7 + 8\,t$$

acceleration

$a_x = 8\ m/s^2$
$y = 5t$
velocity in y direction
$v_y = 5\ m/s$
acceleration
$a_y = 0$
Net acceleration
$a = a_x i + a_y j$
$a = 8\ m/s^2$

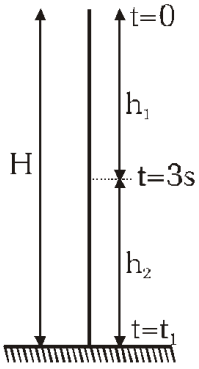
6) As $\vec{u}.\vec{a} < 0$
So initially speed of particle decreases.

7) $2ax = (50^2) - (10)^2$
and $2(-a)(-x) = v^2 - (50)^2$
This gives; $v^2 - (50)^2 = (50)^2 - (10)^2$
i.e. $v = 70 m/s$



8)
$$\left(\frac{V}{3}\right)^2 = V^2 - 2a(4) \Rightarrow a = \frac{8V^2}{9(8)} = \frac{V^2}{9}$$
$$0 = V^2 - 2a(4 + x)$$
$$\Rightarrow \quad V^2 = 2\left(\frac{V^2}{9}\right)(4 + x)$$
$4.5 = 4 + x$
$x = 0.5$

9) $S_1 + S_2 + S_3 = 5S$
$$S + vt + \frac{v^2}{2a'} = 5S$$
$$vt + \frac{v^2}{2\left(\frac{f}{2}\right)} = 4S \qquad \left(\because v = \sqrt{2fs}\right)$$
$$4S = vt + \frac{v^2}{f}$$
$$4S = \sqrt{2fs}\, t + \frac{2fs}{f}$$
$$2S = t\sqrt{2fs}$$
$$S = \frac{1}{2}ft^2$$



10) $H = \frac{1}{2}g(5)^2 = \frac{1}{2}(10)(5)^2 = 125m$

$h_1 = \frac{1}{2}g(3)^2 = 45m$

$h_2 = H - h_1 = 80m$

But $h_2 = \frac{1}{2}g(t_1 - 3)^2$

So, $80 = \frac{1}{2}(10)(t_1 - 3)^3$

$\Rightarrow t_1 - 3 = 4 \Rightarrow t_1 = 7s$

11)

→ In moving horizontal distance 5m, the ball will fall by distance $\frac{1}{2}gt^2$.

→ Time taken by particle to move 5m is,

$$t = \frac{5}{10\cos 37^\circ} = \frac{5}{8}\sec$$

→ Distance travelled by particle from C to D is,

$$y_0 = \frac{1}{2}g\left(\frac{5}{8}\right)^2 = \frac{1}{2} \times 10 \times \frac{25}{64} = \frac{125}{64} = 1.95m$$

12) $v = \frac{ds}{dt}$ and $a = \frac{dv}{dt}$

13)

$y = mx \Rightarrow V^2 = \frac{4}{4}t \quad \Rightarrow V^2 = t \Rightarrow V = \sqrt{t}$

$a = \frac{dV}{dt} = \frac{1}{2\sqrt{t}} = \frac{1}{2\sqrt{4}} = \frac{1}{4}m/s^2$

14) Area of a–x graph

$$= \int a\,dx = \int \left(v\frac{dv}{dx}\right)dx = \int v\,dv$$

$$\Rightarrow \int_0^{v_m} v\,dv = \frac{1}{2} \times 1 \times 1$$

$$\Rightarrow \frac{v_m^2}{2} = \frac{1}{2} \Rightarrow v_m = 1\ m/s$$

15) Here v and a have +ve signs ⇒ **speeding up**

16)

$$\frac{2u_y}{g} = \frac{2u_x u_y}{g} = \frac{\sqrt{3}}{g}$$

$\therefore u_x = 1$ & $u_y = \frac{\sqrt{3}}{2}$

$\therefore \tan\theta = \frac{u_y}{u_x} = \frac{\sqrt{3}}{2} \Rightarrow \theta = \tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

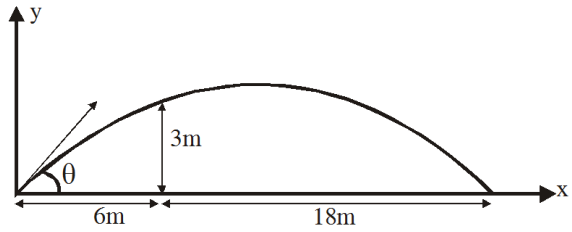
17) $\frac{d\vec{v}}{dt} = \vec{a} = \vec{g}$ = constant

18)

From equation of trajectory, $y = x\tan\theta\left[1 - \frac{x}{R}\right]$

$\Rightarrow 3 = 6\tan\theta\left[1 - \frac{1}{4}\right]$

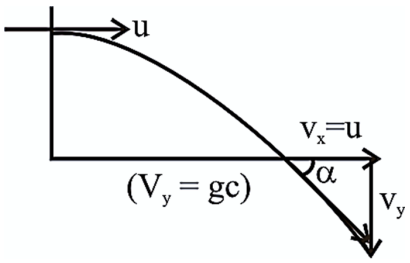
$\Rightarrow \tan\theta = \frac{2}{3}$

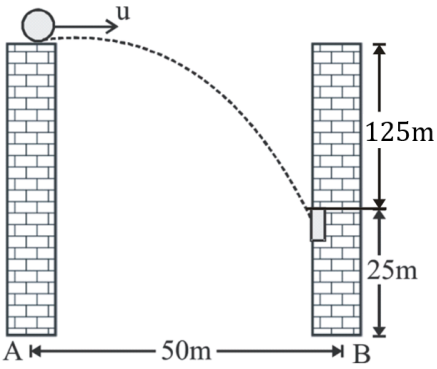


19)

$\tan\alpha = \frac{gt}{u} = \frac{10 \times 10}{500} = \frac{1}{5}$

$\alpha = \tan^{-1}\left(\frac{1}{5}\right)$





20)

$125 = \frac{1}{2}gt^2$

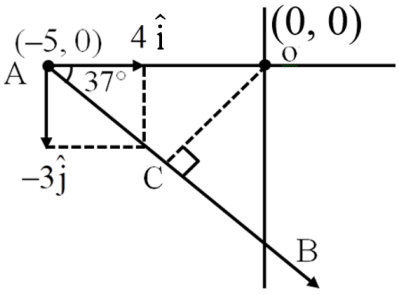
$t^2 = 25$, $t = 5$ sec.

$u \times t = 50$

$u = 10$m/s.

21) Relative acceleration of coin w.r.t. lift = 1m/s^2 ↓

$$t = \sqrt{\frac{2h}{a_{rel}}} = \sqrt{\frac{2 \times 2}{1}} = 2 \text{ s}$$



22) OC = 5 sin 37°

OC = 3 m

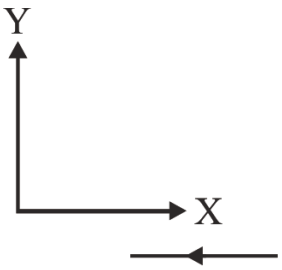
23) Velocity of man is

$$\bar{v}_M = v\cos 60^\circ \hat{i} - v\sin 60^\circ \hat{j} = \frac{v}{2}\hat{i} - \frac{\sqrt{3}}{2}v\hat{j}$$

Velocity of rain w.r.t man is $\bar{v}_{RM} = -v\hat{i}$

or $\bar{v}_R - \bar{v}_M = -\bar{v}\hat{i}$

$$\bar{v}_R = -v\hat{i} + \left(\frac{v}{2}\hat{i} - \frac{\sqrt{3}}{2}v\hat{j}\right)$$

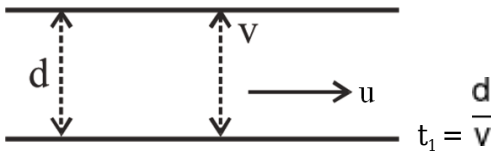


$\vec{v} = -\frac{v}{2}\hat{i} - \frac{\sqrt{3}}{2}v\hat{j}$

$|\vec{v}_R| = \sqrt{\frac{v^2}{4} + \frac{3v^2}{4}} = v$

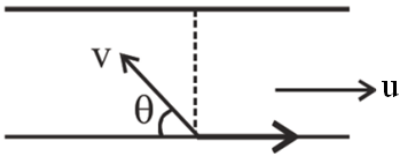
24)

For shortest time :



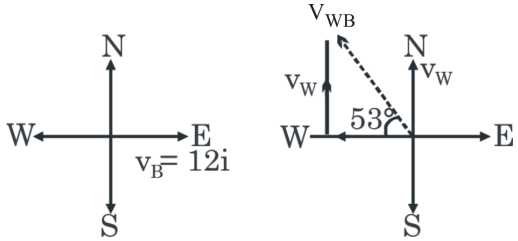
$t_1 = \frac{d}{v}$

For shortest distance :



$t_2 = \frac{d}{v \sin\theta}$

$\frac{t_1}{t_2} = \frac{\frac{d}{v}}{\frac{d}{v\sin\theta}} = \sin\theta$



25)

$v_{WB} = \vec{v}_W - \vec{v}_B$

$\tan 53° = \frac{v_w}{12} = \frac{4}{3}$ $v_w = 16\ ms^{-1}$

26)

Explanation : Five balls interact electrostatically. Given which pairs attract or repel, we must

identify the charge state of ball1.

Concept : Like charges repel, opposite charges attract. Neutral bodies attract charged ones but do not repel any.

Formula :
no formula - purely based on electrostatic interaction logic :
Like charge → Repulsion
Unlike or neutral → Attraction

Calculation :
(1,2) attracts → 1 is opposite or neutral to 2
(2,3) repels → 2 and 3 same charge
(2,4) attracts → 2 and 4 opposite or neutral
(4,5) repels → 4 and 5 same charges as 2
(4,1) attracts → 1 is opposite or neutral to 4
So, 2,3,4,5 are all charged. Since 2 and 4 are oppositely charged and both attract 1,1 cannot be opposite to both so it must be neutral.
Final option (3)

27)

Charge at B is fix

force on charge A ⇒ $f_A = \frac{k(q)(4q)}{(2d)^2} + \frac{k(q)(2q)}{d^2}$

$= \frac{3kq^2}{d^2}$

force on charge C ⇒ $f_C = \frac{k(q)(4q)}{(2d)^2} + \frac{k(2q)(4q)}{d^2}$

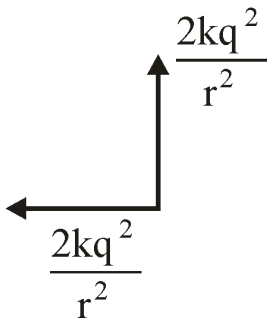
$= \frac{9kq^2}{d^2}$

Tension in string AB ⇒ $T = f_A$

Tension in string BC ⇒ $T' = f_C$

$\frac{T}{T'} = \frac{1}{3}$

28)



$2\sqrt{2}\frac{kq^2}{r^2} = F_{net}$

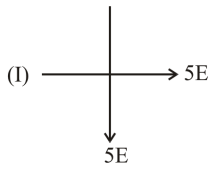
29) $K_i + U_i = K_f + U_f$

30) Equilibrium would be stable, when the particle displaced from equilibrium position comes back to its equilibrium position under the action of restoring forces and equilibrium would be unstable, when particle has no tendency to come back to equilibrium position when displaced from the equilibrium position.

31) $E = -y\hat{i} - x\hat{j}$

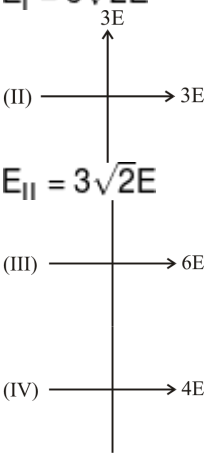
$\tan\theta = \frac{E_y}{E_x} = \frac{x}{y}$

32)



If $E = \frac{kq}{r^2}$

$E_I = 5\sqrt{2}E$



$E_{II} = 3\sqrt{2}E$

so, $E_I > E_{III} > E_{II} > E_{IV}$

33) The sign of flux is positive through right end cap.

34) $\Delta V = \left|\frac{KQ}{3r} - \frac{KQ}{r}\right| = \frac{2KQ}{3r} = V \Rightarrow KQ = \frac{3Vr}{2}$

$E = \frac{KQ}{9r^2} = \frac{3Vr}{2(9r^2)} = \frac{V}{6r}$

35)

Electric field due to an uniformly charged arc $E = \frac{2K\lambda}{r}\sin\frac{\phi}{2}$

$\frac{2K\lambda}{r}\sin\frac{\phi}{2} = \frac{\sqrt{3}\lambda}{4\pi\epsilon_0 r}$

$\frac{2\lambda}{4\pi\epsilon_0 r}\sin\frac{\phi}{2} = \frac{\sqrt{3}\lambda}{4\pi\epsilon_0 r}$

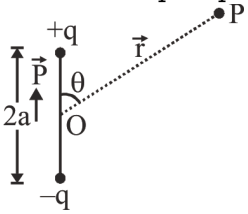
$\sin\frac{\phi}{2} = \frac{\sqrt{3}}{2} = \sin 60° \Rightarrow \frac{\phi}{2} = 60^\circ$

$\phi = 120°$

36) $U = U_1 + U_2 = -pE\cos\theta + \frac{k(q)(-q)}{2\ell}$

$= -pE\cos\theta - \frac{kq^2}{2\ell}$

37) An electric dipole is a pair of equal and opposite charges q and –q separated by some distance 2a. Its dipole moment vector has $\vec{p}$ magnitude 2qa and is in the direction of dipole axis from –q to q.



The electric field due to a dipole at a point P is,

$E = \frac{p}{4\pi\varepsilon_0}\frac{\sqrt{3\cos^2\theta + 1}}{r^3}$, $V = \frac{KP\cos\theta}{r^2}$ (r > > a)

$\vec{\tau} = \vec{p} \times \vec{E}$, $\vec{F}_{Net} = 0$

38) Net $\vec{E}$ inside conductor due to outside charge is zero hence force is also zero.

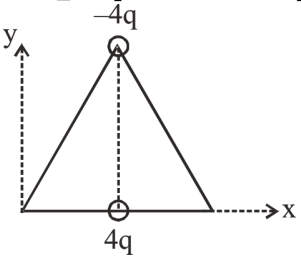
39) $\because E = \frac{\sigma}{\in_0}$; (independent of R if σ is same)

40) $V = \frac{1}{4\pi\varepsilon_0}\frac{q}{R} = \frac{\sigma}{\varepsilon_0}R$

$V_1 = V_2 \Rightarrow \sigma_1 R_1 = \sigma_2 R_2$

$\frac{\sigma_1}{\sigma_2} = \frac{R_2}{R_1}$

41) ∴ dipole moment p = $4q \times \frac{\sqrt{3}}{2}a(-\hat{j})$



42) Potential decreases in the direction of electric field.

43) All options correct

44) **Question Explain:**
We have two concentric spherical conducting shells with charges. The charge on the outer shell is varied while the inner shell's charge remains constant. We need to verify if the potential difference changes.

Concept Based:
The electric Field between the two shells depends only on the charge of the inner shell.

Formula Used:
Electric potential difference between two shells:

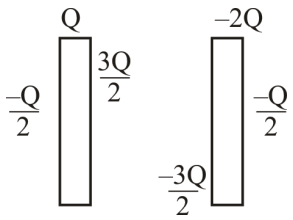
$$V_1 - V_2 = kQ_1\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Where Q_1 is the charge on the inner shell, and R_1, R_2 are the radii of the inner and outer shells.

Calculation:
The potential difference between the shells depends only on the charge of the inner shell (Q_1) and the radii (R_1 and R_2). Since the electric field between the shells depends solely on Q_1, the potential difference does not change when the charge on the outer shell is varied.

Final Answer: (4)
Both Assertion (A) and Reason (R) are correct, and the Reason correctly explains the Assertion.

45)



$$\frac{Q_{sys}}{2} = \frac{Q + (-2Q)}{2} = \frac{-Q}{2}$$